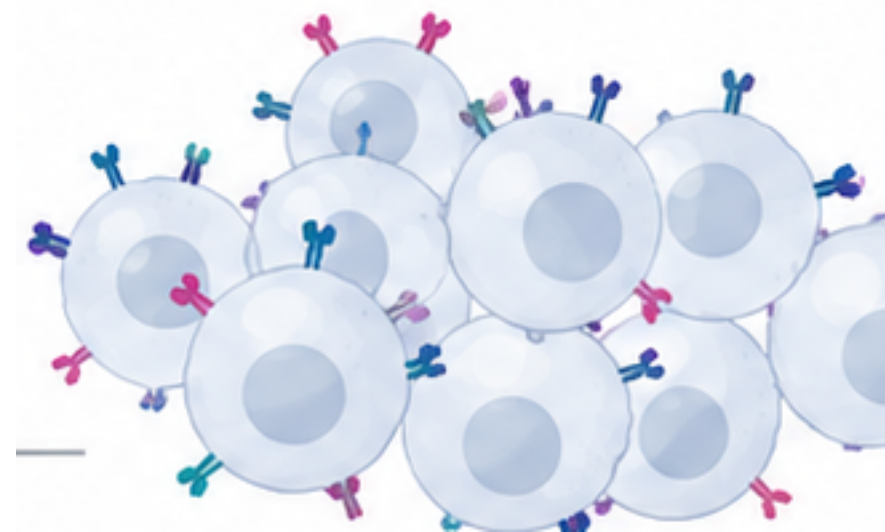


a

Frangieh Perturbo-CITE-seq

RNA + 20 surface proteins in the same 218,331 cells

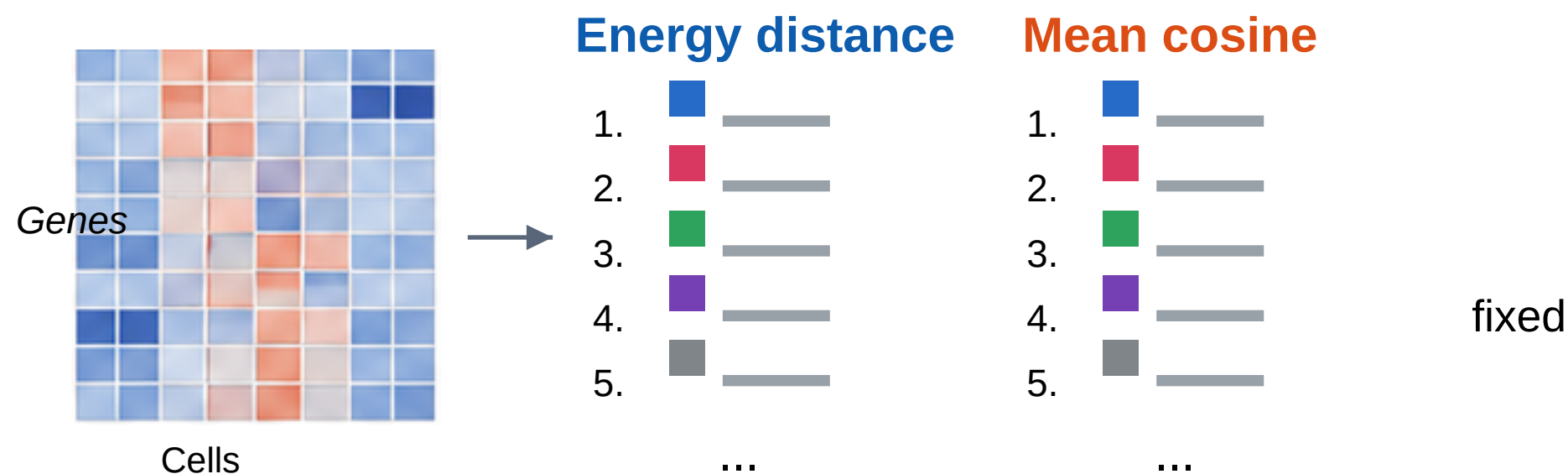
RNA
(transcriptome)



Surface proteins
(20 markers)



RNA-derived rankings

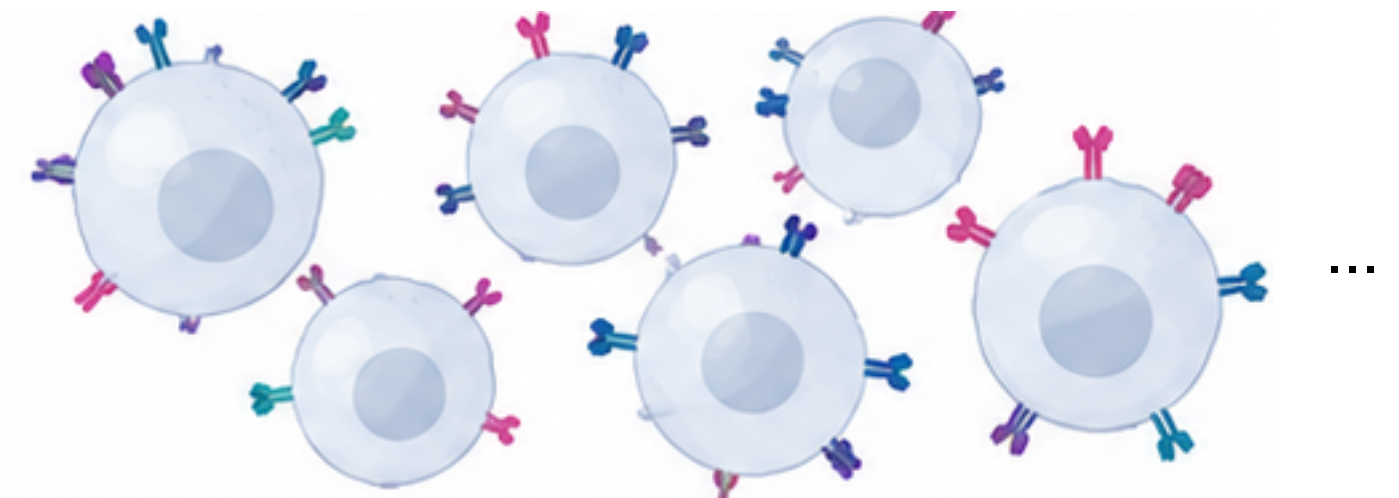


n = 659 queries

only evaluator form changes

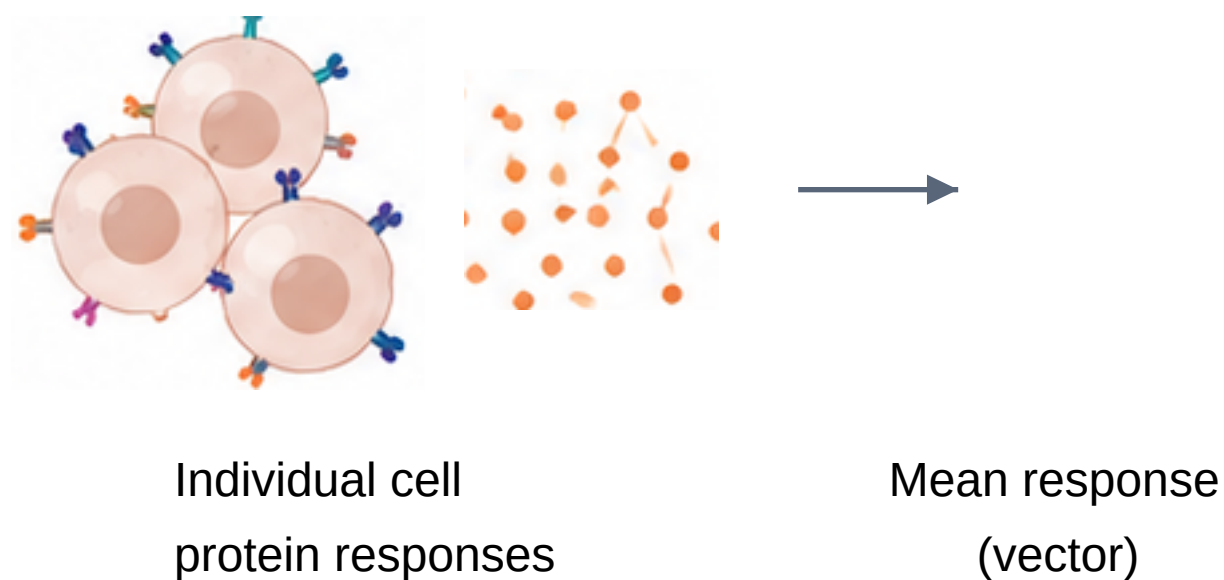
Held-out protein responses

same measurements



Mean-form evaluator

cosine between mean protein responses



Distribution-form evaluator

energy distance between protein-response populations

